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TAG: MV-PA081-R010020	
SERVICE: REACTOR	
ID X TT LENGTH (mm)	4000 X 5100
DESIGN PRESS.MIN / MAX (kg/cm2g)	/ FULL LIQUID
DESIGN TEMP. MIN / MAX (°c)	5.6 / 86
INSULATION / INSULATION THK.(mm)	/
REGENERATION FREQUENCY	NA
TRIM: PA081-BRS-010858-4"-RP109ZDRP-NI	
MOC: IS2026 GR.B0 E250 + HRL	

TAG: MT-PA081-T010021	
SERVICE: RETENTION TANK	
ID X HEIGHT (mm)	7500 X 10550
NOMINAL CAPACITY (m3)	466
DESIGN PRESS.MAX (kg/cm2g)	FULL LIQUID
DESIGN TEMP. MIN / MAX (°C)	5.6 / 86
INSULATION / INSULATION THK. (mm)	PG / 1.2
MOC: IS2026 GR.B0 E250 + HRL	
TRIM: PA081-BRS-010857-4"-RP109ZDRP-NI	

TAG:MF-PA081-A010020
SERVICE:REACTOR AGITATOR
DRIVER RATED POWER (kw): 15.00 kW

TAG:MF-PA081-A000060A/B
SERVICE:FLOCCULANT TANK AGITATOR
DRIVER RATED POWER (kw): 15.00 kW

TAG: MP-PA081-P000060A /B/C /D/E	
SERVICE: FLOCCULANT PUMP	
RATED CAPACITY (m3 /hr)	2
DIFFERENTIAL PRESS. (Kg/cm2g)	3
DRIVER POWER (kW)	1.11
INSULATION / INSULATION THK. (mm)	/
TRIM: RS12EKFR	
MOC: A351 GR. CF8	

TAG: MT-PA081-T000060A/B	
SERVICE: FLOCCULANT TANK	
ID X HEIGHT (mm)	4000 X 5850
NOMINAL CAPACITY (m3)	73.5
DESIGN PRESS.MAX (kg/cm2g)	FULL LIQUID
DESIGN TEMP. MIN / MAX (°C)	5.6 / 60
INSULATION / INSULATION THK. (mm)	/
MOC: A240 TYPE 304	
TRIM: PA081-CH-000840-4"-RS12EKFR-NI	

TAG:MF-PA081-A010021
SERVICE:RETENTION TANK AGITATOR
DRIVER RATED POWER (kw): 90.00 kW

PA081-BRS-011110-6"-RP109ZDRP-NI — SLURRY 000011 379 — FROM CLARIFIER BOTTOM PUMP P010030A/B

PA081-CB-300107-2"-RS12IHFR-NI — 18% CAUSTIC SODA 300001 SHT2 710 — FROM 18% CAUSTIC SODA PUMP P300320A/B

PA081-AP-910680-2"-RC12ABDR-NI — PLANT AIR 900006 711 — FROM HEADER

PA081-BRS-001758-3"-RP1093DRP-NI — BRINE 000017 SHT2 712 — FROM BACKWASH WATER TRANSFER PUMP P000095A/B

RETURN BRINE 4893 100009 — FROM RECOVERY BRINE PUMP P110165A/B — PA081-BRR-110967-6"-RP1092DRP-NI

RETURN BRINE 4898 100011 — FROM RECOVERY BRINE PUMP P120165A/B — PA081-BRR-121167-6"-RP1092DRP-NI

BRINE 698 000005 SHT2 — FROM REACTOR FEED PUMP P000017A/B/C/D/E — PA081-BRS-000506-14"-RP109ZDRP-NI

PA081-BRS-000807-18"-RP109ZDRP-NI

SODA ASH 705 000007 — FROM SODA ASH PUMP P000070 A/B — PA081-CH-000723-2"-RP1092DRP-NI

FROM HEADER 701 900004 SHT2 — DM WATER — PA081-WA-000065-4"-RV12EKFR-NI

PA081-WA-900467-4"-RV12EKFR-NI

PA081-AP-010813-2"-RP109ZDRP-NI

PA081-AP-010805-2"-RP109ZDRP-NI

PA081-AP-010881-2"-RP109ZDRP-NI

PA081-BRS-010808-8"-RP109ZDRP-NI

PA081-BRS-001814-3"-RP109ZDRP-NI

PA081-BRS-010803-24"-FRP GUTTER-NI — BRINE 381 000011 — TO BRINE CLARIFIER Y010030

PA081-BRS-010823-8"-RP109ZDRP-NI

PA081-AP-010812-2"-RP109ZDRP-NI

PA081-BRS-010806-8"-RP109ZDRP-NI

PA081-AP-010810-2"-RP109ZDRP-NI

PA081-AP-010811-2"-RP109ZDRP-NI

PA081-BRS-010801-18"-RP109ZDRP-NI

PA081-BRS-010802-18"-RP109ZDRP-NI

PA081-CH-000835-2"-RS12EKFR-NI

PA081-CH-000823-2"-RS12EKFR-NI

FLOCCULANT 715 000009 — TO STATIC MIXER PZMX-0015

FLOCCULANT 716 000010 — TO STATIC MIXER PZMX-0016

PA081-CH-000834-2"-RS12EKFR-NI

PA081-CH-010822-2"-RS12EKFR-NI

PA081-CH-000824-1/2"-RS12EKFR-NI

PA081-CH-000816-2"-RS12EKFR-NI

PA081-CH-000826-1/2"-RS12EKFR-NI

PA081-CH-000817-2"-RS12EKFR-NI

PA081-CH-000828-1/2"-RS12EKFR-NI

PA081-CH-000818-2"-RS12EKFR-NI

PA081-CH-000830-1"-RS12EKFR-NI

PA081-CH-000819-2"-RS12EKFR-NI

PA081-CH-000832-1"-RS12EKFR-NI

PA081-CH-000820-2"-RS12EKFR-NI

PA081-CH-000825-1"-RS12EKFR-NI

PA081-CH-000827-1"-RS12EKFR-NI

PA081-CH-000829-1"-RS12EKFR-NI

PA081-CH-000831-1"-RS12EKFR-NI

PA081-CH-000833-1"-RS12EKFR-NI

PA081-CH-000812-3"-RS12EKFR-NI

PA081-CH-000813-3"-RS12EKFR-NI

PA081-CH-000814-3"-RS12EKFR-NI

PA081-CH-000851-3"-RS12EKFR-NI

PA081-CH-000815-3"-RS12EKFR-NI

PA081-CH-000811-4"-RS12EKFR-NI

PA081-CH-000810-4"-RS12EKFR-NI

PA081-CH-000855-6"-RS12EKFR-NI

PA081-CH-000856-6"-RS12EKFR-NI

PA081-CH-000854-8"-RS12EKFR-NI

PA081-CH-000853-8"-RS12EKFR-NI

SP PA081-PZMX-0014

SP PA081-PZRN-0001

1R-020
MV-PA081-R010020
REACTOR
1A-020
MF-PA081-A010020
REACTOR AGITATOR
MP-PA081-M010020
REACTOR AGITATOR MOTOR

1D-021
MT-PA081-T010021
RETENTION TANK
1A-021
MF-PA081-A010021
RETENTION TANK AGITATOR
MF-PA081-M010021
RETENTION TANK AGITATOR MOTOR

D-060B
MT-PA081-T000060B
FLOCCULANT TANK
A-060B
MF-PA081-A000060B
FLOCCULANT TANK AGITATOR
MF-PA081-M000060B
FLOCCULANT TANK AGITATOR MOTOR

D-060A
MT-PA081-T000060A
FLOCCULANT TANK
A-060A
MF-PA081-A000060A
FLOCCULANT TANK AGITATOR
MF-PA081-M000060A
FLOCCULANT TANK AGITATOR MOTOR

MV-PA081-V000060A /B/C /D/E
SUCTION PULSATION DAMPENER
MV-PA081-V000060F /G/H /I /J
DISCHARGE PULSATION DAMPENER

MV-PA081-V000060 P/Q/R/S/T
CALIBRATION POT

P-060A/B/C/D/E
MP-PA081-P000060A /B /C /D /E
FLOCCULANT PUMP
MP-PA081-M000060A /B /C /D/E
FLOCCULANT PUMP MOTOR

NOTES

GENERAL NOTES

1. FOR GENRAL NOTES & LEGENDS REFER DRAWINGS GP3201-G11-PA081-050001 TO GP3201-G11-PA081-050008.
2. INSTRMENT TAG NUMBERS ON THIS DRAWING ARE TO BE PREFIXED WTH "081".
3. ELEVATION 100.5M CORRESOPOND TO FGL.
4. DELETED.

NOTES

1. RECTANGULAR SHAPED OPEN GUTTER OF 24" X 24" X 24" SIZ E TO BE PROVIDED.
2. TO BE ROUTED TO NEAREST TRENCH.
3. REFER DRAWING GP3201-G11-PA081-050007 FOR PUMP & AGITATOR MOTOR DETAILS.
4. DEVIATION ALARM SHALL BE GENERATED IN DCS WHEREVER 2 NOS. OF LEVEL TRANSMITTERS ARE PROVIDED ON THE SAME TANK/VESSEL AND THE DIFFERENCE BETWEEN THEIR MEASURED VALUES ARE BEYOND A LIMIT.
5. GUTTER TO BE PROVIDED WITH TWO REMOVABLE COVER/OPENING NEAR INLET & OUTLET RESPECTIVELY. 04
6. 2" NOZZLES AT 120° (3 NOZZLES) FOR AIR AGITATION.
7. SUTAIBLE STRUCTURE TO BE PROVIDED FOR AGITATOR INSTALLATION AND MAINTENANCE.
8. DISSIMILAR TYPE NRV SHALL BE PROVIDED.
9. PROVISION FOR LINE FLUSHING.
10. REFER SELECTOR SWITCH TABLE FOR DETAILS.
11. REFER DETAILED P & ID (G23P05-D1GPA-MHH001-TEM-P-03-081-C02-005) FOR FLOCULANT HANDLING SYSTEM.
12. EMERGENCY FEED HOPPER
13. CALIBRATION POT, SUCTION DAMPNER, DISCHARGE DAMPNER, PSV LOOSE SUPPLY BY PUMP VENDOR. 04

HOLDS

1. CONTROL VALVE AND ITS BYPASS VALVE SIZE UNDER HOLD.
2. SIZE OF EXPANDER & REDUCER ARE UNDER HOLD.
3. DELETED.
4. DELETED.
5. DELETED. 04
6. NOZZLE SIZE SHALL BE CONFIRMED LATER.
7. DELETED.
8. DELETED. 04

ALARM/ INTERLOCK SCHEDULE

INTERLOCK NO	INSTRUMENT NO	INTERLOCK DESCRITION	ACTION
I-000018	LT-000009	AT HIGH HIGH LEVEL	CLOSE XV-000003
I-000019	LT-000009	AT LOW LOW LEVEL 04	TRIP AGITATOR A000060B
I-000020	LT-000009	AT LOW LOW LOW LEVEL 04	TRIP PUMP P000060A/B/C/D/E
I-000021	LT-000010	AT HIGH HIGH LEVEL	CLOSE XV-000002
I-000022	LT-000010	AT LOW LOW LOW LEVEL 04	TRIP PUMP P000060A/B/C/D/E
I-000023	LT-000010	AT LOW LOW LEVEL 04	TRIP AGITATOR A000060A
I-000024	--	PUMP P000017 A & P000017 B NOT RUNNING 04	CLOSE FV000012 AND AV000001 04

SELECTOR SWITCH DESCRIPTION

SR.NO	SWITCH NO.	P&ID	SELECTION	EFFECT
1.	S-000003	GP3201-G41-PA081-000008	SELCTION OF T000060 A/B	ACTIVATION OF ALL ASSOCIATED INTERLOCKS OF T000060A OR T000060B

REV	DATE	REVISION	BY	CHECK	APPD.	HOD	PROJ ENG	PMT
05	07 Oct 25	ISSUED FOR CONSTRUCTION	DMR	TSB	MPB	SN	MPB	MPG
04	28 Mar 25	RE-ISSUED FOR DESIGN	ASK	VKB	MPB	SN	MPB	VRK
03	26 Sep 24	ISSUED FOR DESIGN	RRL	TSB	MPB	SN	MPB	AJ
02	16 Aug 24	ISSUED FOR HAZOP	RRL	TSB	MPB	SN	MPB	AJ
01	20 Jun 24	ISSUED FOR REVIEW	RRL	TSB	MPB	SN	MPB	AJ

TATA CONSULTING ENGINEERS LIMITED
MUMBAI

JOB IDENTIFIER: G23P05 | MOC NO: NA

Reliance
RELIANCE INDUSTRIES LIMITED
DAHEJ MANUFACTURING DIVISION

BECHTEL
PMC : BNT Offshore Solutions, Inc.

PROJECT :
VINYLS PROGRAM
CHLOR ALKALI (1035 KTA CHLORINE)

DRAWING TITLE:
PIPING AND INSTRUMENTATION DIAGRAM
PRIMARY BRINE PURIFICATION
(REACTOR, RETENTION & FLOCCULANT TANK)

SCALE	PROJECT NO.	DRAWING NO.	REV.
NTS	GP3201	GP3201-G41-PA081-000008	04